FMCW/TWTT Annotated Source Atlas

84-source map for the Saeed AWR2944 timing project

Summer 2026

This atlas is the source-by-source companion to the literature review, mathematical handbook, and MATLAB simulation blueprint. It records why each source is in the corpus, what to extract from it, whether a physical research copy is present, and where it lives in the archive. It is deliberately exhaustive so the literature repository remains navigable after the first simulation sprint.

Contents

1 How to use this atlas

The atlas is not a substitute for reading the priority papers. It is a navigation and extraction map. For every paper that becomes active in the project, record (1) waveform/architecture, (2) unknown parameters, (3) estimator, (4) required synchronization/coherence assumptions, (5) demonstrated precision/accuracy, and (6) one limitation that matters for two AWR2944 boards.

The priority labels are working labels: P0 is foundational/immediate, P1 is strong second-wave material, P2 is useful supporting literature, and "reference" denotes documentation or context. "Citation only" means the source remains in the research map but its PDF bytes are not present in the current archive.

2 P0 - 77 GHz cooperative FMCW

1. A 77-GHz Cooperative Radar System Based on Multi-Channel FMCW Stations for Local Positioning Applications (2013)

Priority/status P0 / `citation_only`
Authors R. Feger et al.
DOI / identifier `10.1109/TMTT.2012.2227781`
Archive path not present
Why it is here. Important bridge to 77-GHz hardware.

What to extract for this project. Use to understand which observables survive separate oscillators, what degree of coherence is required, and how cross-path data should be retained/combined across nodes.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

3 P0 - FMCW fundamentals

2. Linear FMCW Radar Techniques (1992)

Priority/status P0 / `citation_only`
Authors A. G. Stove
DOI / identifier `10.1049/ip-f-2.1992.0048`
Archive path not present
Why it is here. Classic reference.

What to extract for this project. Reference for FMCW/radar fundamentals or adjacent methods. Use it to sanity-check notation, signal relationships, and assumptions when its topic becomes active in the implementation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

4 P0 - MATLAB / simulation

3. A Versatile FMCW Radar System Simulator for Millimeter-Wave Applications (2008)

Priority/status P0 / `included_restricted_user_supplied`
Authors S. Scheiblhofer; M. Treml; S. Schuster; R. Feger; A. Stelzer
DOI / identifier `10.1109/EUMC.2008.4751778`
Archive path `09_RESTRICTED_OR_CITATION_ONLY/user_supplied_restricted/2008_Scheiblhofer_A_Versatile_FMCW_Radar_System_Simulator.pdf`

Why it is here. Modular MATLAB simulator from sweep synthesis through baseband.

What to extract for this project. Use to define simulator architecture, signal-domain fidelity, validation strategy, or efficient baseband implementation. Extract module boundaries, required state variables, and validation metrics.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

5 P0 - TWTT / cooperative FMCW

4. Method for High Precision Radar Distance Measurement and Synchronization of Wireless Units (2007)

Priority/status P0 / `included_restricted_user_supplied`
Authors S. Roehr; M. Vossiek; P. Gulden
DOI / identifier `10.1109/MWSYM.2007.380436`
Archive path `09_RESTRICTED_OR_CITATION_ONLY/user_supplied_restricted/2007_Roehr_High_Precision_Radar_Distance_Synchronization.pdf`

Why it is here. Core predecessor; <100 ps time synchronization reported.

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

5. Coherent Full-Duplex Double-Sided Two-Way Ranging and Velocity Measurement Between Separate Incoherent Radio Units (2019)

Priority/status P0 / `included_restricted_user_supplied`
Authors M. Gottinger; F. Kirsch; P. Gulden; M. Vossiek
DOI / identifier `10.1109/TMTT.2019.2902553`
Archive path `09_RESTRICTED_OR_CITATION_ONLY/user_supplied_restricted/2019_Gottinger_CFDDS_TWR.pdf`

Why it is here. Closest academic analogue: reciprocal FMCW with separate clocks.

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

6 P0 - clock synchronization

6. Method for High Precision Clock Synchronization in Wireless Systems with Application to Radio Navigation (2007)

Priority/status P0 / `citation_only`
Authors S. Roehr; P. Gulden; M. Vossiek
DOI / identifier `10.1109/RWS.2007.351890`
Archive path not present

Why it is here. Foundational clock synchronization; full copy not physically present in this runtime.

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

7 P0 - cooperative FMCW

7. Precise Distance Measurement with Cooperative FMCW Radar Units (2008)

Priority/status P0 / `citation_only`
Authors A. Stelzer; M. Jahn; S. Scheiblhofer
DOI / identifier `10.1109/RWS.2008.4463606`
Archive path not present
Why it is here. Core cooperative FMCW ranging paper.

What to extract for this project. Use to understand which observables survive separate oscillators, what degree of coherence is required, and how cross-path data should be retained/combined across nodes.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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8. Precise Distance and Velocity Measurement for Real Time Locating in Multipath Environments Using a Frequency-Modulated Continuous-Wave Secondary Radar Approach (2008)

Priority/status P0 / `citation_only`
Authors S. Roehr; P. Gulden; M. Vossiek
DOI / identifier `10.1109/TMTT.2008.2003137`
Archive path not present
Why it is here. Detailed range/velocity/multipath model.

What to extract for this project. Use to understand which observables survive separate oscillators, what degree of coherence is required, and how cross-path data should be retained/combined across nodes.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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8 P0 - frequency/phase estimation

9. Accuracy Limits of a K-Band FMCW Radar with Phase Evaluation (2012)

Priority/status P0 / `citation_only`
Authors S. Scherr; S. Ayhan; M. Pauli; T. Zwick
DOI / identifier not recorded
Archive path not present
Why it is here. CRLB and phase-evaluation precision precedent.

What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards.* This classification is more valuable than a generic summary.

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10. An Efficient Frequency and Phase Estimation Algorithm With CRB Performance for FMCW Radar Applications (2015)

Priority/status P0 / `included_restricted_user_supplied`
Authors S. Scherr et al.
DOI / identifier `10.1109/TIM.2014.2381354`
Archive path `09_RESTRICTED_OR_CITATION_ONLY/user_supplied_restricted/2015_Scherr_Efficient_Frequency_and_Phase_Estimation_CRB.pdf`
Why it is here. CZT plus phase evaluation; CRB context.

What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards.* This classification is more valuable than a generic summary.

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9 P0 - multistatic synchronization

11. FMCW Radar Transceiver Synchronization in a Multistatic Microwave Tomography System (2023)

Priority/status	P0 / `citation_only`
Authors	B. Lettner
DOI / identifier	not recorded
Archive path	not present

Why it is here. Thesis-length synchronization treatment.

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

10 P0 - nonidealities

12. Impact of Frequency Ramp Nonlinearity, Phase Noise, and SNR on FMCW Radar Accuracy (2016)

Priority/status P0 / `included_restricted_user_supplied`
Authors S. Ayhan et al.
DOI / identifier `10.1109/TMTT.2016.2599165`
Archive path `09_RESTRICTED_OR_CITATION_ONLY/user_supplied_restricted/2016_Ayhan_Ramp_Nonlinearity_Phase_Noise_SNR.pdf`

Why it is here. Analytical and measured accuracy effects.

What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

11 P0 - phase noise

13. The Effect of Phase Noise on Ranging Uncertainty in FMCW Secondary Radar-Based Local Positioning Systems (2012)

Priority/status P0 / `citation_only`
Authors R. Ebelt; D. Shmakov; M. Vossiek
DOI / identifier not recorded
Archive path not present
Why it is here. Directly relevant to secondary/cooperative FMCW PN.

What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

12 P0 - phase noise / PLL

14. Analysis of Ranging Precision in an FMCW Radar Measurement Using a Phase-Locked Loop (2018)

Priority/status	P0 / `included_restricted_user_supplied`
Authors	F. Herzel; D. Kissinger; H. J. Ng
DOI / identifier	`10.1109/TCSI.2017.2733041`
Archive path	`09_RESTRICTED_OR_CITATION_ONLY/user_supplied_restricted/2018_Herzel_Ranging_Precision_FMCW_PLL.pdf`

Why it is here. PLL phase noise plus receiver noise precision model.

What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

13 P0 - phase noise / nonidealities

15. Detailed Analysis and Modeling of Phase Noise and Systematic Phase Distortions in FMCW Radar Systems (2022)

Priority/status	P0 / `included_restricted_user_supplied`
Authors	P. Tschapek et al.
DOI / identifier	`10.1109/JMW.2022.3195574`
Archive path	`09_RESTRICTED_OR_CITATION_ONLY/user_supplied_restricted/2022_Tschapek_Detailed_Phase_Noise_Systematic_Distortions.pdf`

Why it is here. Colored PN plus systematic phase-error Monte Carlo model.

What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

14 P0 - precision / phase noise

16. Performance Analysis of Cooperative FMCW Radar Distance Measurement Systems (2008)

Priority/status P0 / `citation_only`
Authors S. Scheiblhofer et al.
DOI / identifier `10.1109/MWSYM.2008.4633118`
Archive path not present
Why it is here. Phase-noise performance comparison; web-accessible metadata, PDF not copied.

What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

15 P1 - coded / multistatic synchronization

17. On the Synchronization of Uncoupled Multistatic PMCW Radars (2024)

Priority/status P1 / `citation_only`
Authors D. Werbunat et al.
DOI / identifier `10.1109/TMTT.2024.3359035`
Archive path not present

Why it is here. Digital carrier/timing recovery analog for future coded FMCW.

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

16 P1 - coded FMCW

18. Phase-Coded FMCW Automotive Radar: System Design and Interference Mitigation (2020)

Priority/status P1 / `included_acquired_research_copy`
Authors F. Uysal
DOI / identifier `10.1109/TVT.2019.2953305`
Archive path `04_CODED_FMCW_AND_MULTIRADAR/additional_acquired/2020_Uysal_Phase_Coded_FMCW_System_Design_Interference_Mitigation.pdf`

Why it is here. Accepted manuscript known at TU Delft; not copied due repository download restriction.

What to extract for this project. Use only after the uncoded V0/V1 model is validated. Extract how waveform identity is embedded, how synchronization/code alignment is performed, and whether dechirp preserves the low-bandwidth receiver advantage.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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19. System Level Synchronization of Phase-Coded FMCW Automotive Radars for RadCom (2020)

Priority/status P1 / `included_restricted_user_supplied`
Authors F. Lampel et al.
DOI / identifier `10.23919/EuCAP48036.2020.9135417`
Archive path `09_RESTRICTED_OR_CITATION_ONLY/user_supplied_restricted/2020_Lampel_System_Level_Synchronization_Phase_Coded_FMCW_RadCom.pdf`

Why it is here. Phase-coded FMCW synchronization / RadCom.

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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17 P1 - distributed coherent radar

20. Coherent Signal Processing for Loosely Coupled Bistatic Radar (2021)

Priority/status P1 / `included_restricted_user_supplied`
Authors M. Gottinger; P. Gulden; M. Vossiek
DOI / identifier `10.1109/TAES.2021.3050650`
Archive path `09_RESTRICTED_OR_CITATION_ONLY/user_supplied_restricted/2021_Gottinger_Loosely_Coupled_Bistatic_Radar.pdf`

Why it is here. Coherent processing despite only loose coupling.

What to extract for this project. Use to understand which observables survive separate oscillators, what degree of coherence is required, and how cross-path data should be retained/combined across nodes.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

18 P1 - radar network signal model

21. Signal Model for Coherent Processing of Uncoupled and Low Frequency Coupled MIMO Radar Networks (2024)

Priority/status P1 / `citation_only`
Authors V. Janoudi et al.
DOI / identifier `10.1109/JMW.2023.3334757`
Archive path not present
Why it is here. Strong modern nonideal signal model.

What to extract for this project. Use to map the abstract model to AWR2944/DCA1000 configuration fields, timing windows, sample rates, clocking, calibration, channel topology, and raw-data interpretation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

19 P1 - radar network synchronization

22. Over-the-Air Synchronization for Coherent Digital Automotive Radar Networks (2024)

Priority/status	P1 / `citation_only`
Authors	L. Sigg et al.
DOI / identifier	`10.1109/TRS.2024.3449333`
Archive path	not present

Why it is here. CFO/SFO/timing correction.

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

20 P1 - radar networks

23. Coherent Automotive Radar Networks: The Next Generation of Radar-Based Imaging and Mapping (2021)

Priority/status P1 / `included_restricted_user_supplied`
Authors M. Gottinger et al.
DOI / identifier `10.1109/JMW.2020.3034475`
Archive path `09_RESTRICTED_OR_CITATION_ONLY/user_supplied_restricted/2021_Gottinger_Coherent_Automotive_Radar_Networks.pdf`

Why it is here. Distributed radar classification and 76-77 GHz experiments.

What to extract for this project. Use to understand which observables survive separate oscillators, what degree of coherence is required, and how cross-path data should be retained/combined across nodes.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

21 P2 - additional acquired

24. 2019 MathWorks Automotive Radar Development MATLAB (2019)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `02_MATLAB_AND_SIMULATION/additional_acquired/2019_MathWorks_Automotive_Radar_Development_MATLAB.pdf`

Why it is here. Acquired during the literature sweep and included as a research copy; see original source for licensing/redistribution terms.

What to extract for this project. Use to define simulator architecture, signal-domain fidelity, validation strategy, or efficient baseband implementation. Extract module boundaries, required state variables, and validation metrics.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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25. 2022 Duerr Coherent Multistatic MIMO Radar Network Phase Noise Optimized Synthesis (2022)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `04_CODED_FMCW_AND_MULTIRADAR/additional_acquired/2022_Duerr_Coherent_Multistatic_MIMO_Radar_Network_Phase_Noise_Optimized_Synthesis.pdf`

Why it is here. Acquired during the literature sweep and included as a research copy; see original source for licensing/redistribution terms.

What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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26. 2024 Ahmadi et al Distributed Massive MIMO FMCW Radar Simulator Ray Tracing (2024)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `02_MATLAB_AND_SIMULATION/additional_acquired/2024_Ahmadi_et_al_Distributed_Massive_MIMO_FMCW_Radar_Simulator_Ray_Tracing.pdf`

Why it is here. Acquired during the literature sweep and included as a research copy; see original source for licensing/redistribution terms.

What to extract for this project. Use to define simulator architecture, signal-domain fidelity, validation strategy, or efficient baseband implementation. Extract module boundaries, required state variables, and validation metrics.

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22 P2 - extended bibliography

27. 2014 Frischen Hasch Waldschmidt - Cooperative Radar: Uncorrelated Phase Noise (year n/a)

Priority/status	P2 / `included_acquired_research_copy`
Authors	not recorded in compact catalog
DOI / identifier	not recorded
Archive path	`04_CODED_FMCW_AND_MULTIRADAR/additional_acquired/2014_Frischen_Hasch_Waldschmidt_Cooperative_Radar_Uncorrelated_Phase_Noise.pdf`

Why it is here. Listed in prior acquisition staging; no separate PDF bytes available in this runtime. Physical research copy is included in the topical archive; check original publication license before redistribution.

What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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28. 2014 Rajan van der Veen - Joint Ranging and Synchronization for Anchorless Mobile Networks (year n/a)

Priority/status	P2 / `included_acquired_research_copy`
Authors	not recorded in compact catalog
DOI / identifier	not recorded
Archive path	`03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/2014_Rajan_van_der_Veen_Joint_Ranging_Synchronization_Anchorless_Mobile_Network.pdf`

Why it is here. Listed in prior acquisition staging; no separate PDF bytes available in this runtime. Physical research copy is included in the topical archive; check original publication license before redistribution.

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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29. 2015 Dwivedi - Joint Ranging and Clock Parameter Estimation via Two-Way RTT (year n/a)

Priority/status	P2 / `included_acquired_research_copy`
Authors	not recorded in compact catalog
DOI / identifier	not recorded
Archive path	`03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/2015_Dwivedi_Joint_Ranging_Clock_Parameter_Estimation_Two_Way_RTT.pdf`

Why it is here. Listed in prior acquisition staging; no separate PDF bytes available in this runtime. Physical research copy is included in the topical archive; check original publication license before redistribution.

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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30. 2015 ITU - Two-Way Satellite Time and Frequency Transfer (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/2015_ITU_TF1153_Two_Way_Satellite_Time_Frequency_Transfer.pdf`

Why it is here. Listed in prior acquisition staging; no separate PDF bytes available in this runtime. Physical research copy is included in the topical archive; check original publication license before redistribution.

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

31. 2016 Lopez-Martinez et al. - Simulation of FMCW Radar with SDR (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `02_MATLAB_AND_SIMULATION/additional_acquired/2016_Lopez_Martinez_Vidal_Morera_Simulation_FMCW_Radar_SDR.pdf`

Why it is here. Listed in prior acquisition staging; no separate PDF bytes available in this runtime. Physical research copy is included in the topical archive; check original publication license before redistribution.

What to extract for this project. Use to define simulator architecture, signal-domain fidelity, validation strategy, or efficient baseband implementation. Extract module boundaries, required state variables, and validation metrics.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

32. 2016 Scheiblhofer - In-Chirp FSK Communication Between Cooperative 77-GHz Radar Stations (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `04_CODED_FMCW_AND_MULTIRADAR/additional_acquired/2016_Scheiblhofer_In_Chirp_FSK_Communication_Cooperative_77GHz_Radar.pdf`

Why it is here. Listed in prior acquisition staging; no separate PDF bytes available in this runtime. Physical research copy is included in the topical archive; check original publication license before redistribution.

What to extract for this project. Use only after the uncoded V0/V1 model is validated. Extract how waveform identity is embedded, how synchronization/code alignment is performed, and whether dechirp preserves the low-bandwidth receiver advantage.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

33. 2017 TI - Complex Baseband Architecture in FMCW Radar Systems (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `01_FMCW_FUNDAMENTALS/classic_and_signal_processing/2017_TI_Complex_Baseband_Architecture_in_FMCW_Radar_Systems.pdf`

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What to extract for this project. Reference for FMCW/radar fundamentals or adjacent methods. Use it to sanity-check notation, signal relationships, and assumptions when its topic becomes active in the implementation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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34. 2018 Kazaz - Joint Ranging and Clock Synchronization for Dense IoT (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/2018_Kazaz_Joint_Ranging_Clock_Synchronization_Dense_IoT.pdf`

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What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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35. 2018 Park et al. - Leakage Mitigation / Internal Delay Compensation in FMCW (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `05_PRECISION_LIMITS_AND_NONIDEALITIES/additional_acquired/2018_Park_Park_Park_Leakage_Mitigation_Internal_Delay_Compensation_FMCW.pdf`

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What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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36. 2018 Svensson - High Resolution Frequency Estimation for FMCW Radar (thesis) (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `05_PRECISION_LIMITS_AND_NONIDEALITIES/additional_acquired/2018_Svensson_High_Resolution_Frequency_Estimation_FMCW_Radar_Thesis.pdf`

Why it is here. Listed in prior acquisition staging; no separate PDF bytes available in this runtime. Physical research copy is included in the topical archive; check original publication license before redistribution.

What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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37. 2019 Mueller Diewald - Cooperative Radar Signature Method for Unambiguity (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `04_CODED_FMCW_AND_MULTIRADAR/additional_acquired/2019_Mueller_Diewald_Cooperative_Radar_Signature_Unambiguity.pdf`

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What to extract for this project. Use only after the uncoded V0/V1 model is validated. Extract how waveform identity is embedded, how synchronization/code alignment is performed, and whether dechirp preserves the low-bandwidth receiver advantage.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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38. 2019 Wang et al. - Range Accuracy of FMCW Radar with Source Nonlinearity (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `05_PRECISION_LIMITS_AND_NONIDEALITIES/additional_acquired/2019_Wang_et_al_Range_Accuracy_FMCW_Source_Nonlinearity.pdf`

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What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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39. 2020 Infineon - Phase Coded FMCW Radar patent (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `04_CODED_FMCW_AND_MULTIRADAR/additional_acquired/2020_Infineon_US20200132825A1_Phase_Coded_FMCW_Radar.pdf`

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What to extract for this project. Use only after the uncoded V0/V1 model is validated. Extract how waveform identity is embedded, how synchronization/code alignment is performed, and whether dechirp preserves the low-bandwidth receiver advantage.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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40. 2020 UIUC - FMCW Radar lecture (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `02_MATLAB_AND_SIMULATION/additional_acquired/2020_UIUC_FMCW_Radar_Lecture.pdf`

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What to extract for this project. Reference for FMCW/radar fundamentals or adjacent methods. Use it to sanity-check notation, signal relationships, and assumptions when its topic becomes active in the implementation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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41. 2020 Uysal Orru - Phase-Coded FMCW Automotive Radar: Application and Challenges (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `04_CODED_FMCW_AND_MULTIRADAR/additional_acquired/2020_Uysal_Orru_Phase_Coded_FMCW_Application_and_Challenges.pdf`

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What to extract for this project. Use only after the uncoded V0/V1 model is validated. Extract how waveform identity is embedded, how synchronization/code alignment is performed, and whether dechirp preserves the low-bandwidth receiver advantage.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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42. 2022 Architectures and Synchronization Techniques for Distributed Satellite Systems - Survey (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/2022_Architectures_and_Synchronization_Techniques_Distributed_Satellite_Systems_Survey.pdf`

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What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards.* This classification is more valuable than a generic summary.

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43. 2022 Duerr - Coherent Multistatic MIMO Radar Network with Phase-Noise-Optimized Synthesis (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `06_TI_AWR2944_IMPLEMENTATION/chirp_config_calibration_and_MIMO/TI_MIMO_Radar.pdf`

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What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards.* This classification is more valuable than a generic summary.

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44. 2022 MIT - Radar and Coherent Sensor Processing lecture (year n/a)

Priority/status P2 / `included_acquired_research_copy`
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DOI / identifier not recorded
Archive path `02_MATLAB_AND_SIMULATION/additional_acquired/2022_MIT_Radar_and_Coherent_Sensor_Processing_Lecture.pdf`

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What to extract for this project. Use to understand which observables survive separate oscillators, what degree of coherence is required, and how cross-path data should be retained/combined across nodes.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards.* This classification is more valuable than a generic summary.

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45. 2022 TI - Interference Mitigation for AWR/IWR Devices (year n/a)

Priority/status	P2 / `included_acquired_research_copy`
Authors	not recorded in compact catalog
DOI / identifier	not recorded
Archive path	`01_FMCW_FUNDAMENTALS/classic_and_signal_processing/2022_TI_Interference_Mitigation_for_AWR_IWR_Devices.pdf`

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What to extract for this project. Reference for FMCW/radar fundamentals or adjacent methods. Use it to sanity-check notation, signal relationships, and assumptions when its topic becomes active in the implementation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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46. 2023 Tagliaferri et al. - Cooperative Coherent Multistatic Imaging and Phase Synchronization (year n/a)

Priority/status	P2 / `included_acquired_research_copy`
Authors	not recorded in compact catalog
DOI / identifier	not recorded
Archive path	`03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/2023_Tagliaferri_et_al_Cooperative_Coherent_Multistatic_Imaging_Phase_Synchronization.pdf`

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What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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47. 2024 Ahmadi et al. - Distributed Massive MIMO FMCW Radar Simulator with Ray Tracing (year n/a)

Priority/status	P2 / `included_acquired_research_copy`
Authors	not recorded in compact catalog
DOI / identifier	not recorded
Archive path	`06_TI_AWR2944_IMPLEMENTATION/chirp_config_calibration_and_MIMO/TI_MIMO_Radar.pdf`

Why it is here. Listed in prior acquisition staging; no separate PDF bytes available in this runtime. Physical research copy is included in the topical archive; check original publication license before redistribution.

What to extract for this project. Use to define simulator architecture, signal-domain fidelity, validation strategy, or efficient baseband implementation. Extract module boundaries, required state variables, and validation metrics.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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48. 2024 Chen Niknejad - RF-Domain Leakage Cancellation for FMCW Radars (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `05_PRECISION_LIMITS_AND_NONIDEALITIES/additional_acquired/2024_Chen_Niknejad_RF_Domain_Leakage_Cancellation_FMCW_Radars.pdf`

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What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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49. 2024 Huang et al. - Overview of Millimeter-Wave Radar Modeling Methods (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `02_MATLAB_AND_SIMULATION/additional_acquired/2024_Huang_et_al_Overview_Millimeter_Wave_Radar_Modeling_Methods.pdf`

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What to extract for this project. Use to define simulator architecture, signal-domain fidelity, validation strategy, or efficient baseband implementation. Extract module boundaries, required state variables, and validation metrics.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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50. 2024 Kou et al. - Distributed PMCW Radar Network Phase Noise (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `04_CODED_FMCW_AND_MULTIRADAR/additional_acquired/2024_Kou_Bauduin_Bourdoux_Pollin_Distributed_PMCW_Radar_Network_Phase_Noise.pdf`

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What to extract for this project. Use only after the uncoded V0/V1 model is validated. Extract how waveform identity is embedded, how synchronization/code alignment is performed, and whether dechirp preserves the low-bandwidth receiver advantage.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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51. 2024 Multi-Objective Distributed Beamforming with High-Accuracy Synchronization (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/2024_Multi_Objective_Distributed_Beamforming_High_Accuracy_Sync.pdf`

Why it is here. Listed in prior acquisition staging; no separate PDF bytes available in this runtime. Physical research copy is included in the topical archive; check original publication license before redistribution.

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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52. 2025 BIPM - Time Dissemination/TWSTFT lecture (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
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Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/2025_BIPM_Time_Dissemination_TWSTFT_Lecture.pdf`

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What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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53. 2025 DLR - Investigation of CW/LFM Waveforms for Bi/Multistatic Radar Synchronization (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/2025_DLR_Investigation_CW_LFM_Waveforms_Bi_Multistatic_Radar_Synchronization.pdf`

Why it is here. Listed in prior acquisition staging; no separate PDF bytes available in this runtime. Physical research copy is included in the topical archive; check original publication license before redistribution.

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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54. 2025 Eid - Novel Simulation Technique for UWB FMCW Radar (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `02_MATLAB_AND_SIMULATION/additional_acquired/2025_Eid_Novel_Simulation_Technique_UWB_FMCW_Radar.pdf`

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What to extract for this project. Use to define simulator architecture, signal-domain fidelity, validation strategy, or efficient baseband implementation. Extract module boundaries, required state variables, and validation metrics.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards.* This classification is more valuable than a generic summary.

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55. 2025 Han Meng Masouros - OTA Time/Frequency Synchronization for Distributed ISAC (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/2025_Han_Meng_Masouros_OTA_Time_Frequency_Synchronization_Distributed_ISAC.pdf`

Why it is here. Listed in prior acquisition staging; no separate PDF bytes available in this runtime. Physical research copy is included in the topical archive; check original publication license before redistribution.

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards.* This classification is more valuable than a generic summary.

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56. 2025 Multistatic Radar Performance with Distributed Wireless Synchronization / BCRLB (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/2025_Multistatic_Radar_Performance_Distributed_Wireless_Synchronization_BCRLB.pdf`

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What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards.* This classification is more valuable than a generic summary.

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57. 2025 Real-Time High-Accuracy Digital Wireless Time/Frequency/Phase Synchronization (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/2025_Real_Time_High_Accuracy_Digital_Wireless_Time_Frequency_Phase_Synchronization.pdf`

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What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards.* This classification is more valuable than a generic summary.

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58. 2026 UCSB - Introduction to mmWave Radar Sensing Lab Notes (year n/a)

Priority/status P2 / `included_acquired_research_copy`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `02_MATLAB_AND_SIMULATION/additional_acquired/2026_UCSB_Introduction_to_mmWave_Radar_Sensing_Lab_Notes.pdf`

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What to extract for this project. Use to map the abstract model to AWR2944/DCA1000 configuration fields, timing windows, sample rates, clocking, calibration, channel topology, and raw-data interpretation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards.* This classification is more valuable than a generic summary.

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59. Merlo - Picosecond NLOS Wireless Time/Frequency Transfer presentation (year n/a)

Priority/status P2 / `included_acquired_research_copy`
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DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/Merlo_Picosecond_NLOS_Wireless_Time_Frequency_Transfer_Presentation.pdf`

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What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards.* This classification is more valuable than a generic summary.

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23 reference - source documentation

60. 2013 Pfeffer FMCW MIMO Frequency Division TX Beamforming (year n/a)

Priority/status reference / `included_restricted_user_supplied`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `09_RESTRICTED_OR_CITATION_ONLY/user_supplied_restricted/2013_Pfeffer_FMCW_MIMO_Frequency_Division_TX_Beamforming.pdf`
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What to extract for this project. Reference for FMCW/radar fundamentals or adjacent methods. Use it to sanity-check notation, signal relationships, and assumptions when its topic becomes active in the implementation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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61. 2022 Kumbul Smoothed Phase Coded FMCW (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `04_CODED_FMCW_AND_MULTIRADAR/phase_coded_FMCW/2022_Kumbul_Smoothed_Phase_Coded_FMCW.pdf`
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What to extract for this project. Use only after the uncoded V0/V1 model is validated. Extract how waveform identity is embedded, how synchronization/code alignment is performed, and whether dechirp preserves the low-bandwidth receiver advantage.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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62. 2022 Merlo Mghabghab Nanzer Wireless Picosecond Time Synchronization (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/additional_acquired/2022_Merlo_Mghabghab_Nanzer_Wireless_Picosecond_Time_Synchronization.pdf`
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What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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63. 2023 Merlo High Accuracy Wireless Time Frequency Transfer Distributed Beamforming (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/picosecond_wireless_sync/2023_Merlo_High_Accuracy_Wireless_Time_Frequency_Transfer_Distributed_Beamforming.pdf`

Why it is here. source_or_downloaded

What to extract for this project. Use for the precision/error-budget layer. Implement the impairment before the estimator whenever possible, then compare bias/variance/RMSE with the paper's analytical or measured behavior.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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64. 2024 Decentralized Picosecond Synchronization Distributed Wireless Systems (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/picosecond_wireless_sync/2024_Decentralized_Picosecond_Synchronization_Distributed_Wireless_Systems.pdf`

Why it is here. source_or_downloaded

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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65. 2024 Shandi Merlo Nanzer Wireless Picosecond Time Synchronization Dynamic Connectivity (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/picosecond_wireless_sync/2024_Shandi_Merlo_Nanzer_Wireless_Picosecond_Time_Synchronization_Dynamic_Connectivity.pdf`

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What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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66. 2026-08-07 Saeed Meeting Notes (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `07_PROJECT_CONTEXT/2026-08-07_Saeed_Meeting_Notes.pdf`
Why it is here. source_or_downloaded

What to extract for this project. Reference for FMCW/radar fundamentals or adjacent methods. Use it to sanity-check notation, signal relationships, and assumptions when its topic becomes active in the implementation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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67. BIPM TWSTFT Calibration Guidelines 2016 (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/foundations_and_standards/BIPM_TWSTFT_Calibration_Guidelines_2016.pdf`
Why it is here. source_or_downloaded

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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68. FMCW TWTT acquisition status 2026-08-10.md (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path not present
Why it is here. source_or_downloaded

What to extract for this project. Reference for FMCW/radar fundamentals or adjacent methods. Use it to sanity-check notation, signal relationships, and assumptions when its topic becomes active in the implementation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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69. Honeywell EP2985625A1 FMCW Radar with Timing Synchronization (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/patents/Honeywell_EP2985625A1_FMCW_Radar_with_Timing_Synchronization.pdf`
Why it is here. source_or_downloaded

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

70. Honeywell US20160047892A1 FMCW Radar Phase Encoded Data Channel (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `04_CODED_FMCW_AND_MULTIRADAR/phase_coded_FMCW/Honeywell_US20160047892A1_FMCW_Radar_Phase_Encoded_Data_Channel.pdf`
Why it is here. source_or_downloaded

What to extract for this project. Reference for FMCW/radar fundamentals or adjacent methods. Use it to sanity-check notation, signal relationships, and assumptions when its topic becomes active in the implementation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

71. MathWorks Design of FMCW Radars for Active Safety Applications (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `02_MATLAB_AND_SIMULATION/MathWorks/MathWorks_Design_of_FMCW_Radars_for_Active_Safety_Applications.pdf`
Why it is here. source_or_downloaded

What to extract for this project. Reference for FMCW/radar fundamentals or adjacent methods. Use it to sanity-check notation, signal relationships, and assumptions when its topic becomes active in the implementation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

72. MathWorks Radar System Design Using MATLAB Simulink 2016 (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `02_MATLAB_AND_SIMULATION/MathWorks/MathWorks_Radar_System_Design_Using_MATLAB_Simulink_2016.pdf`
Why it is here. source_or_downloaded

What to extract for this project. Use to define simulator architecture, signal-domain fidelity, validation strategy, or efficient baseband implementation. Extract module boundaries, required state variables, and validation metrics.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

73. NIST Fundamentals of Two Way Time Transfers by Satellite (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `03_TWO_WAY_TIME_TRANSFER_AND_SYNC/foundations_and_standards/NIST_Fundamentals_of_Two_Way_Time_Transfers_by_Satellite.pdf`
Why it is here. source_or_downloaded

What to extract for this project. Use to formalize clock epoch offset, clock-rate/frequency offset, reciprocity, two-way cancellation, presynchronization, and calibration. Translate each distinct clock/path error into an explicit simulator parameter.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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74. TI AWR2943 AWR2944 Datasheet RevE 2026 (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `06_TI_AWR2944_IMPLEMENTATION/AWR2944/TI_AWR2943_AWR2944_Datasheet_RevE_2026.pdf`
Why it is here. source_or_downloaded

What to extract for this project. Use to map the abstract model to AWR2944/DCA1000 configuration fields, timing windows, sample rates, clocking, calibration, channel topology, and raw-data interpretation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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75. TI AWR2944 AWR2944P EVM User Guide RevC (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `06_TI_AWR2944_IMPLEMENTATION/AWR2944/TI_AWR2944_AWR2944P_EVM_User_Guide_RevC.pdf`
Why it is here. source_or_downloaded

What to extract for this project. Use to map the abstract model to AWR2944/DCA1000 configuration fields, timing windows, sample rates, clocking, calibration, channel topology, and raw-data interpretation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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76. TI AWR294x Device Errata RevC (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `06_TI_AWR2944_IMPLEMENTATION/AWR2944/TI_AWR294x_Device_Errata_RevC.pdf`
Why it is here. source_or_downloaded

What to extract for this project. Use to map the abstract model to AWR2944/DCA1000 configuration fields, timing windows, sample rates, clocking, calibration, channel topology, and raw-data interpretation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

77. TI AWR294x Technical Reference Manual RevD (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `06_TI_AWR2944_IMPLEMENTATION/AWR2944/TI_AWR294x_Technical_Reference_Manual_RevD.pdf`

Why it is here. source_or_downloaded

What to extract for this project. Use to map the abstract model to AWR2944/DCA1000 configuration fields, timing windows, sample rates, clocking, calibration, channel topology, and raw-data interpretation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

78. TI Cascade Coherency Phase Shifter Calibration (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `06_TI_AWR2944_IMPLEMENTATION/chirp_config_calibration_and_MIMO/TI_Cascade_Coherency_Phase_Shifter_Calibration.pdf`

Why it is here. source_or_downloaded

What to extract for this project. Use to map the abstract model to AWR2944/DCA1000 configuration fields, timing windows, sample rates, clocking, calibration, channel topology, and raw-data interpretation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

79. TI Fundamentals of mmWave Radar Sensors (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `01_FMCW_FUNDAMENTALS/TI_and_intro/TI_Fundamentals_of_mmWave_Radar_Sensors.pdf`

Why it is here. source_or_downloaded

What to extract for this project. Use to map the abstract model to AWR2944/DCA1000 configuration fields, timing windows, sample rates, clocking, calibration, channel topology, and raw-data interpretation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

80. TI Getting Started with mmWave Sensors 2025 (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `01_FMCW_FUNDAMENTALS/TI_and_intro/TI_Getting_Started_with_mmWave_Sensors_2025.pdf`

Why it is here. source_or_downloaded

What to extract for this project. Use to map the abstract model to AWR2944/DCA1000 configuration fields, timing windows, sample rates, clocking, calibration, channel topology, and raw-data interpretation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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81. TI MIMO Radar (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `06_TI_AWR2944_IMPLEMENTATION/chirp_config_calibration_and_MIMO/TI_MIMO_Radar.pdf`

Why it is here. source_or_downloaded

What to extract for this project. Use to map the abstract model to AWR2944/DCA1000 configuration fields, timing windows, sample rates, clocking, calibration, channel topology, and raw-data interpretation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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82. TI Programming Chirp Parameters (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `06_TI_AWR2944_IMPLEMENTATION/chirp_config_calibration_and_MIMO/TI_Programming_Chirp_Parameters.pdf`

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What to extract for this project. Use to map the abstract model to AWR2944/DCA1000 configuration fields, timing windows, sample rates, clocking, calibration, channel topology, and raw-data interpretation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

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83. TI Self Calibration mmWave Radar Devices (year n/a)

Priority/status reference / `included_topical`
Authors not recorded in compact catalog
DOI / identifier not recorded
Archive path `06_TI_AWR2944_IMPLEMENTATION/chirp_config_calibration_and_MIMO/TI_Self_Calibration_mmWave_Radar_Devices.pdf`

Why it is here. source_or_downloaded

What to extract for this project. Use to map the abstract model to AWR2944/DCA1000 configuration fields, timing windows, sample rates, clocking, calibration, channel topology, and raw-data interpretation.

Notebook prompt. Write down the paper's assumptions in equation form. Mark each assumption as *true in V0*, *true only after hardware synchronization*, *requires calibration*, or *not true for two independent AWR boards*. This classification is more valuable than a generic summary.

84. TI mmWave Sensor Raw Data Capture DCA1000 mmWave Studio (year n/a)

Priority/status	reference / `included_topical`
Authors	not recorded in compact catalog
DOI / identifier	not recorded
Archive path	`06_TI_AWR2944_IMPLEMENTATION/DCA1000_and_capture/TI_mmWave_Sensor_Raw_Data_Capture_DCA1000_mmWave_Studio.pdf`

Why it is here. source_or_downloaded

What to extract for this project. Use to map the abstract model to AWR2944/DCA1000 configuration fields, timing windows, sample rates, clocking, calibration, channel topology, and raw-data interpretation.

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24 Cross-cutting extraction checklist

When the MATLAB model grows, use this checklist to decide which source class to reopen.

Question	Source class to consult
Why does the delayed chirp become a tone?	FMCW fundamentals and simulator papers.
How do time offset and CFO appear together?	2007 cooperative synchronization lineage.
Why can two directions cancel oscillator terms?	TWTT standards and CFDDS-TWR.
Why is an FFT bin not the precision limit?	Scherr/CRLB/frequency-phase estimation literature.
How should phase noise be synthesized?	Herzel, Ayhan, Tschapek, range-correlation literature.
What is a realistic chirp nonlinearity model?	Ayhan, Wang, Tschapek, PLL/ramp papers.
How do we represent independent nodes?	Cooperative/coherent radar-network and modern synchronization papers.
How do we identify multiple chirps/users?	Lampel/Uysal/Kumbul/PMCW/orthogonal-waveform papers.
Which AWR fields set B, T_c, S, F_s, N?	TI chirp-programming, datasheet, TRM, EVM, DCA1000 material.
What does 10 ps really require?	Precision/error-bound + calibration + picosecond wireless time-transfer sources together.

25 Stop condition for the literature phase

The literature phase is sufficiently mature for V0/V1 when every term in the ideal two-way model has a source and a unit test. It is sufficiently mature for a 10-ps claim only when the model contains, or explicitly bounds, clock-rate error, carrier-frequency offset, phase noise with realistic color/correlation, chirp nonlinearity, receiver/ADC noise, fixed and asymmetric group delays, estimator bias/variance, and calibration uncertainty. The latter is a later milestone; it should not delay the first Saeed plots.